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P

COMMON QUARTERLY EXAMINATION - 2025

Standard - 12 Reg.No. | | | | | | |

Time Allowed:3.00 Hours **PHYSICS** *Maximum Marks:70*

PART-I

Note : i) Answer all the questions. **15×1=15**

ii) Chose the most appropriate answer and write the option code and the corresponding answer.

1) A toaster operating at 240 V has a resistance of 120 Ω. Its power is
a) 400 W b) 2 W c) 480 W d) 240 W

2) A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about its axis with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is
a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$

3) Two points A and B are maintained at a potential of 7 V and −4 V respectively. The work done in moving 50 electrons from A to B is
a) 8.80×10^{-17} J b) -8.80×10^{-17} J c) 4.40×10^{-17} J d) 5.80×10^{-17} J

4) When the current changes from +2A to −2A in 0.05 s, an emf of 8V is induced in a coil. The co-efficient of self-induction of the coil is
a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H

5) The electric and magnetic fields of an electromagnetic wave are
a) in phase and perpendicular to each other
b) out of phase and not perpendicular to each other
c) in phase and not perpendicular to each other
d) out of phase and perpendicular to each other

6) The transverse nature of light is shown in,
a) interference b) diffraction c) scattering d) polarisation

7) Stars twinkle due to,
a) reflection b) total internal reflection
c) refraction d) polarisation

8) Which charge configuration produces a uniform electric field?
a) point charge b) uniformly charged infinite line
c) uniformly charged infinite plane d) uniformly charged spherical shell

9) Magnetic field at any point at a distance R due to a long straight conductor carrying current varies as
a) R^2 b) R c) $\frac{1}{R^2}$ d) $\frac{1}{R}$

10) An e.m. wave is propagating in a medium with a velocity $\vec{V} = \vec{V}_i$. The instantaneous oscillating electric field of this e.m. wave is along +y-axis, then the direction of oscillating magnetic field of the e.m. wave will be along:
a) −y direction b) −x direction c) +z direction d) −z direction

11) Resonant angular frequency
a) $\omega r = \frac{1}{\sqrt{LC}}$ b) $\omega r L = \frac{1}{\omega r C}$ c) $X_L = X_C$ d) All of the above

12) A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly
a) 1.0 A m^2 b) 1.2 A m^2 c) 0.5 A m^2 d) 0.8 A m^2

13) In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
a) $\frac{Q}{2}$ b) $\frac{Q}{\sqrt{3}}$ c) $\frac{Q}{\sqrt{2}}$ d) Q

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14) Light transmitted by Nicol prism is,
a) partially polarised b) unpolarised
c) plane polarised d) elliptically polarized
15) The speed of light in an isotropic medium depends on,
a) its intensity b) its wavelength
c) the nature of propagation
d) the motion of the source w.r.t medium

PART-II

Note : Answer any six questions. Question number 24 is compulsory: **6×2=12**

16) What is corona discharge?
17) Distinguish between drift velocity and mobility.
18) Is in ammeter connected in series or parallel in a circuit? why?
19) State Fleming's right hand rule.
20) What is displacement current?
21) Write the two conditions for total internal reflection.
22) State Huygen's principle.
23) Calculate the ratio of electric potential due to point charge q at a point when is at a distance r to the electric potential at a point when the distance is doubled.
24) The resistance of a nichrome wire at 20°C is 10 Ω. If its temperature coefficient of resistivity of nichrome is 0.008/°C. Find the resistance of the wire at boiling point of water. Comment on the result.

PART-III

Note : Answer any six questions. Question number 33 is compulsory: **6×3=18**

25) What are Fraunhofer lines? How are they useful in the identification of element present in the sun?
26) Mention the various energy losses in a transformer.
27) Compare the properties of dia, para and ferro magnetism.
28) Derive an expression for the torque experienced by a dipole due to a uniform electric field.
29) State the applications of seeback effect.
30) Obtain the equation for apparent depth.
31) Write down the equation for a sinusoidal voltage of 50 Hz and its peak value is 20 V.
32) List the properties of electric field lines.
33) Light travelling through transparent oil enters into glass of refractive index 1.5. If the refractive index of glass with respect to the oil is 1.25. What is the refractive of the oil?

PART-IV

Note : Answer all the questions: **5×5=25**

34) a) Derive an expression for electrostatic potential due to an electric dipole. **(OR)**
b) Derive the mirror equation.
35) a) Obtain the condition for bridge balance in Wheatstone's bridge. **(OR)**
b) Obtain the equation for bandwidth in Young's double slit experiment.
36) a) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law. **(OR)**
b) Explain in detail the construction and working of a Van de Graaff generator.
37) a) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit. **(OR)**
b) Explain the determination of the internal resistance of a cell using voltmeter.
38) a) i) The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity of the medium is 2.25. Compute the refractive index of the medium.
ii) Give two uses each of
(i) IR radiation (ii) Microwaves and (iii) UV radiation **(OR)**
b) i) An electron moving perpendicular to a uniform magnetic field 0.500 T undergoes circular motion of radius 2.50 mm. What is the speed of electron?
ii) State and explain Biot-Savart's law.

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